Maryland Crime Data Analysis (1975 - 2020)

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EXECUTIVE SUMMARY

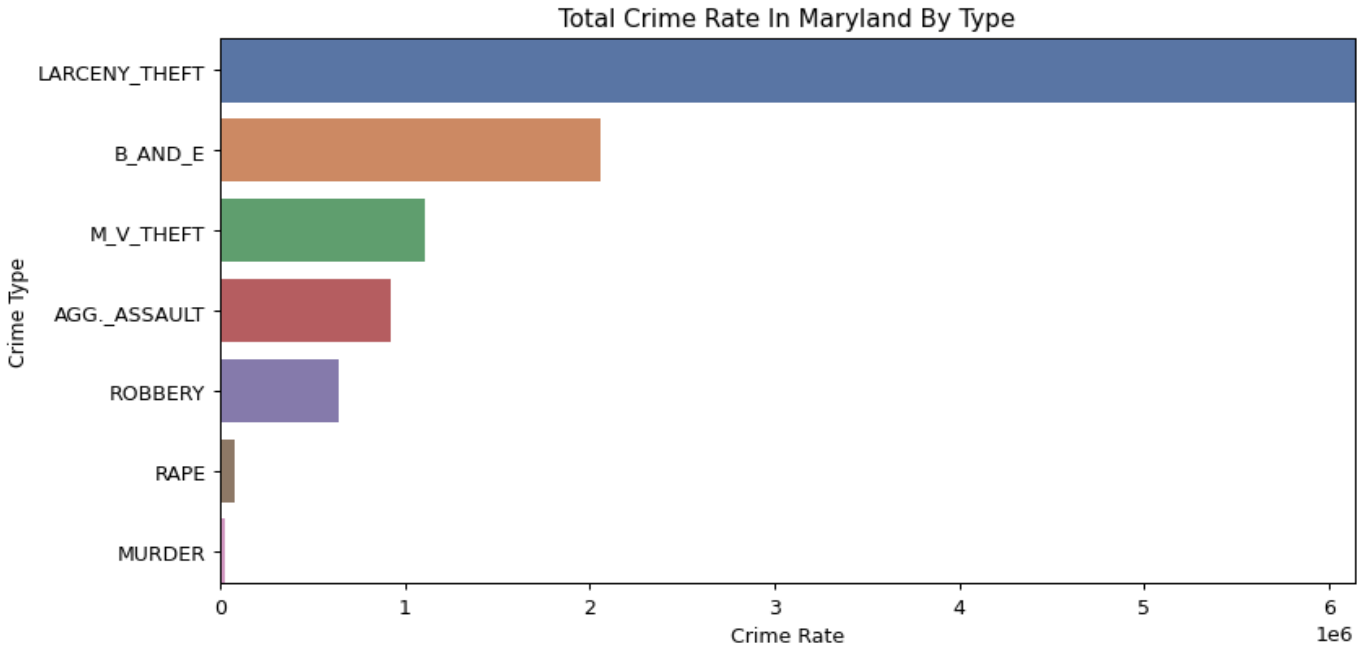
Objective: Identify Maryland crime patterns to assist in resource allocation.

Data Health: High-quality dataset with 1,104 records; minor missing values (approx. 2%) in percent change columns handled via mean-aggregation.

MOST COMMON CRIME TYPES (VOLUME)

- **Rank 1:** Larceny Theft (~6.1 Million total incidents).
- **Rank 2:** Burglary/B&E (~2.06 Million).
- **Rank 3:** Motor Vehicle Theft (~1.1 Million).

Implication: Property crimes vastly outnumber violent crimes in Maryland, suggesting a need for broad-based community surveillance and theft prevention.

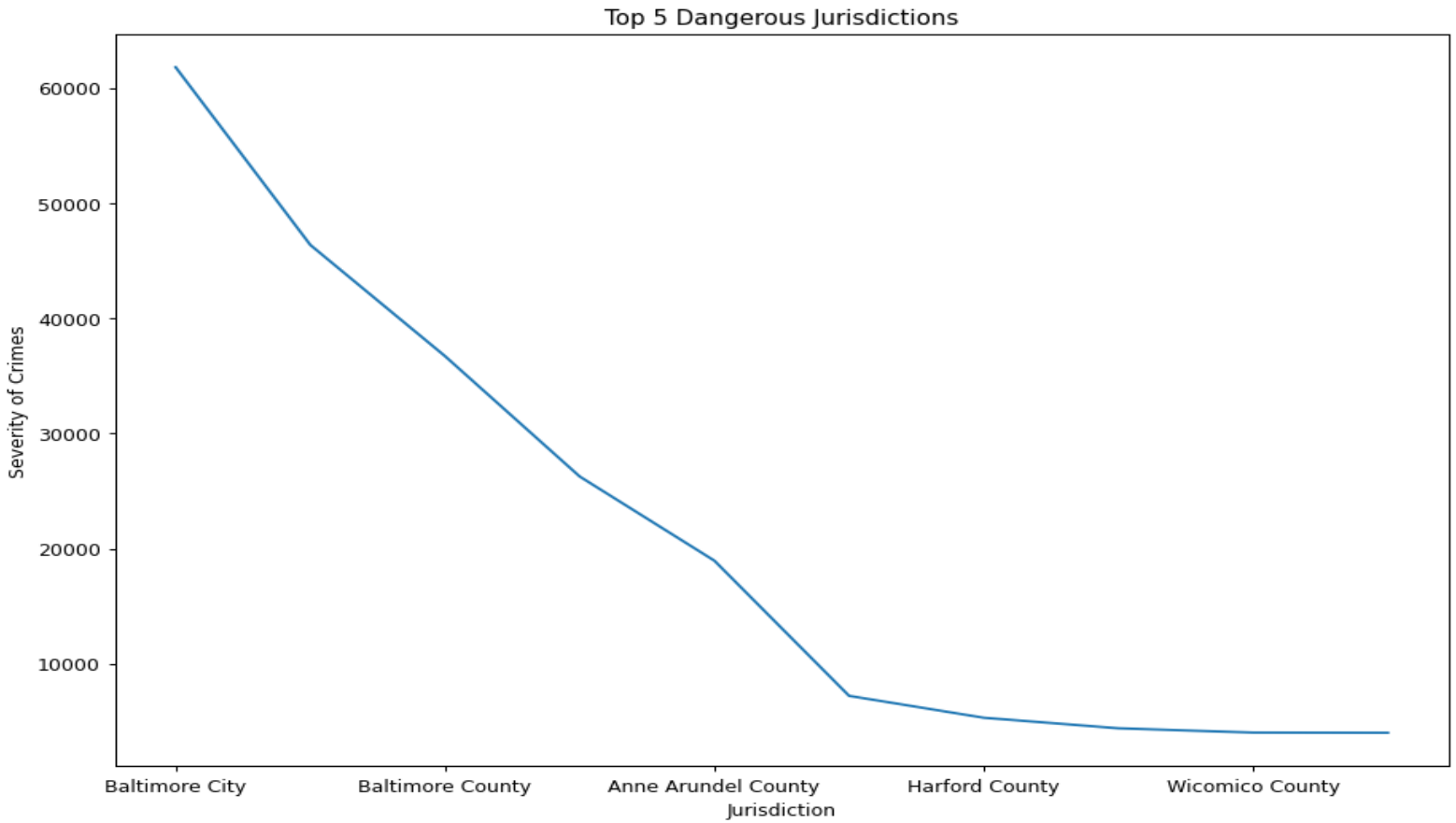

Total Crime Rate In Maryland By Type
LARCENY_THEFT
B_AND_E
M_V_THEFT
AGG._ASSAULT
ROBBERY
RAPE
MURDER
Crime Type
0
1
2
3
4
5
6
1e6
Crime Rate

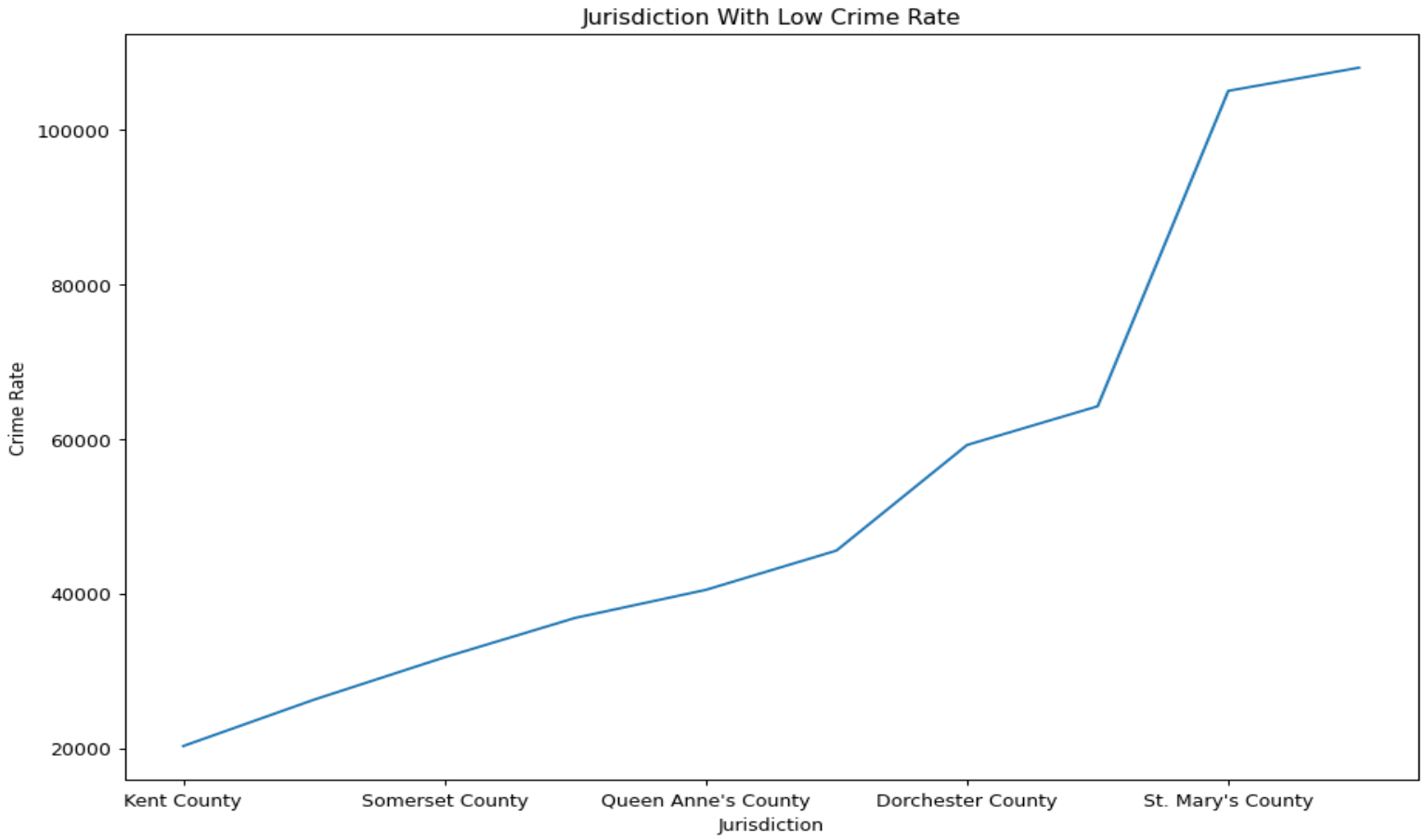
HOTSPOT IDENTIFICATION (2020 DATA)

- **By Volume:**
  - Baltimore City (28,147)
  - Baltimore County (20,045)
  - Prince George's County (19,401)

- **By Rate (per 100k):**
  - Baltimore City (4,782.1)
  - Dorchester County (3,093.7)
  - Worcester County (2,943.0)

Implication: Baltimore City remains the critical priority area, leading in both raw numbers and risk-per-resident.


Top 5 Dangerous Jurisdictions
Severity of Crimes
60000
50000
40000
30000
20000
10000
Baltimore City
Baltimore County
Anne Arundel County
Harford County
Wicomico County
Jurisdiction

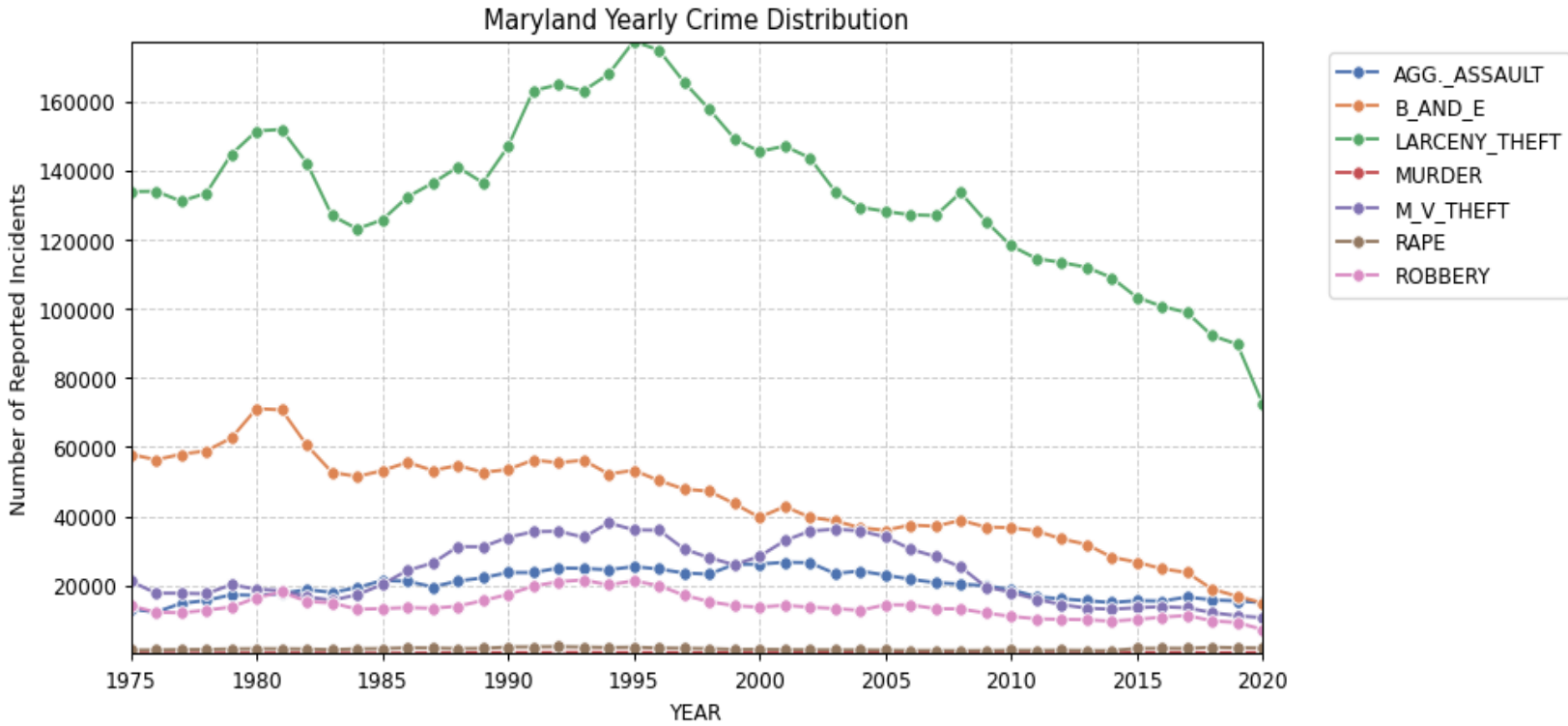

Jurisdiction With Low Crime Rate
Crime Rate
100000
80000
60000
40000
20000
Kent County
Somerset County
Queen Anne's County
Dorchester County
St. Mary's County
Jurisdiction

POPULATION VS. CRIME RATE

- **Total Volume:** Strong positive correlation. As expected, more people = more crime.

- **Crime Rate:** The smaller jurisdictions like Worcester and Dorchester have higher "danger levels" per person than many high-population suburban counties.

Implication: Policing resources should not just follow population; they must address high-rate rural and seasonal zones.


Maryland Yearly Crime Distribution
Number of Reported Incidents
160000
140000
120000
100000
80000
60000
40000
20000
1975
1980
1985
1990
1995
2000
2005
2010
2015
2020
YEAR
AGG._ASSAULT
B_AND_E
LARCENY_THEFT
MURDER
M_V_THEFT
RAPE
ROBBERY

RECOMMENDATION

Strategic Resource Allocation (The "80/20" Rule)

- **Tier 1 Priority (Volume-Based):** Since the "Big Five" (Baltimore City, Prince George's, Baltimore County, Montgomery, and Anne Arundel) account for over 75% of all incidents, the state should maintain a **centralized Resource Hub**. This hub would provide specialized forensics and investigative support that smaller counties cannot afford.

- **Tier 2 Priority (Rate-Based):** High-rate rural areas like **Dorchester** and **Allegany**(Cumberland) require targeted **Socio-Economic Intervention**. High crime rates in these lower-population areas are often symptoms of the opioid crisis or unemployment; resources here should shift toward "Community-Oriented Policing" and addiction recovery services rather than just "boots on the ground."

RECOMMENDATION II

Specialized Crime Task Forces

- **Larceny & Property Crime Initiative:** As **Larceny Theft** is the number 1 crime by a massive margin (over 6 million incidents), Maryland should implement a statewide **Retail & Auto Theft Task Force**. This includes promoting GPS-tracking for high-value items and improving lighting/surveillance in high-volume shopping districts in Montgomery and Anne Arundel counties.

- **Seasonal "Pop-Up" Policing:** For **Worcester County**, the state should provide **Temporary Duty (TDY) Officers** during the summer months (May–September) to handle the tourism influx without overtaxing the small local permanent police force.

RECOMMENDATION III

Data-Driven Policy Improvements

- **Addressing Volatility:** "Rate Percent Change" volatility in categories like **Murder**. Because a single incident in a small county can cause a 100% spike, the state should use a **3-Year Rolling Average** for funding decisions rather than single-year data to avoid "knee-jerk" budget reactions.

- **Multicollinearity Monitoring:** Since population and total crime are so highly correlated ($r > 0.9$), future models should focus on **Residual Analysis**—looking at jurisdictions that have *more* or *less* crime than their population would predict. Areas with "Negative Residuals" (less crime than expected) should be studied as "Success Models" for the rest of the state.

RECOMMENDATION IV

Public Safety Technology

- **Modernization of Reporting:** Given the 1,104 entries across decades, Maryland should move toward **Real-Time Dashboarding**. Providing the public with the "Rate per 100k" (as you calculated) instead of just "Total Counts" will help residents have a more accurate understanding of their actual safety risk.

Thank you !!!